PROFINITE GROUPS

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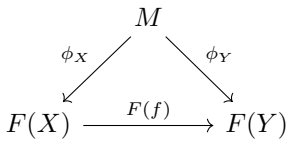
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1. Preliminaries

1.1. **Limits and Inverse Limits.** There are two equivalent definitions for the (category theoretic) limit. One involving cones and the other involving natural transformations. We give the cone definition and leave the other as an exercise.

For a functor $F : \mathcal{J} \to \mathcal{C}$, the image of $\mathcal{J}$ is called a diagram of shape $\mathcal{J}$. Informally, $\mathcal{J}$ can be thought of as an "index category". A **cone** (M, ϕ) to F is an object M of $\mathcal{C}$ along with a family of morphisms $\phi_X : M \to F(X)$, for each X in $\mathcal{J}$, such that the following diagram commutes for each $f : X \to Y$.

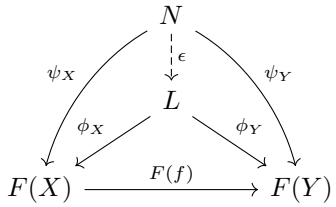


Defining a limit is now easy; a limit is a universal cone. Formally,

Definition 1.1. The **limit** of a functor $F : \mathcal{J} \to \mathcal{C}$ is a cone (L, ϕ) to F such that for any other cone (N, ψ) to F, there is a *unique* morphism $\epsilon : N \to M$ such that

Date: Revised November 4, 2026.

$\psi_X = \phi_X \circ \epsilon$ for all X in $\mathcal{J}$.



We denote the limit of the functor F by $L = \lim F$.

Intuitively, the limit construction subsumes the entire diagram and all its information into a single object $\lim F$ as far as the morphisms into the diagram are concerned. We illustrate this with an example. Further, it generalises many constructions such as the product, pullbacks, etc.

Consider a poset category on the set $(C, \leqslant)$. Let $S \subset C$ be any subset. We aim to find $\lim S$ (abusing notation for the inclusion functor). In this category, a cone from M to S is essentially an upper bound of S since M has a morphism to each of the elements of S. Now, $\lim S$ is an object of C such that any upper bound has a unique morphism to it, i.e. it is an upper bound of S that is smaller than all other upper bounds of S. Sound familiar? Indeed, $\lim S = \sup S$.

We leave the alternate definition of the limit as an exercise.

Exercise 1.1. For an object M of $\mathcal{C}$, define the constant functor $c_M : \mathcal{J} \to \mathcal{C}$ where all objects map to M and all morphisms map to id_M. Show that a cone from M to F is a natural transformation from c_M to F.

Exercise 1.2. Show that $L = \lim F$ if there is a natural transformation $\eta : c_L \to F$ such that any other natural transformation $\epsilon : c_M \to F$ factors through η.

We are mainly concerned with inverse limits of topological groups, so we first develop the theory of topological inverse limits. The reader should be warned that *inverse limits* are limits and *direct limits* are colimits.

We call a partially ordered set I **directed** if any two elements have a common upper bound. More precisely, for each $x, y \in I$ there is $z \in I$ such that $x, y \leqslant z$.

An **inverse system** of topological spaces is a contravariant functor $F : \mathcal{I} \to \mathbf{Top}$ where $\mathcal{I}$ is the poset category on the set I. Expanding this definition out, an inverse system is a collection of topological spaces $\{X_\alpha\}_{\alpha \in I}$ with continuous maps $f_{\alpha\beta} : X_\beta \to X_\alpha$ whenever $\alpha \leqslant \beta \in I$.

Definition 1.2. Let I be a directed set and let $(X_\alpha, f_{\alpha\beta})$ be an inverse system of topological spaces with index set I. The inverse limit of the above system is the limit of the contravariant functor $F : \mathcal{I} \to \mathbf{Top}$ associated to the inverse system.

We denote the inverse limit by $\varprojlim_{\alpha \in I} X_\alpha$ and we shall drop the subscript when there is no risk of confusion. The canonical projections will be denoted by $\pi_\beta : \varprojlim X_\alpha \to X_\beta$ for each $\beta \in I$.

The question of existence and nonemptiness still remains. We prove below that the inverse limit can be constructed as a subspace of a product. Secondly, inverse limits need not always be nonempty but in the case that we are interested in, the spaces are sufficiently nice and inverse limits are, in fact, nonempty.

Proposition 1.3. *Let I be a directed set and let $(X_\alpha, f_{\alpha\beta})$ be an inverse system of topological spaces with index set I. Then the inverse limit exists and is given by*

$$\varprojlim_{\alpha\in I} X_\alpha = \left\{ x \in \prod_{\alpha\in I} X_\alpha \;\middle|\; \pi_\alpha(x) = f_{\alpha\beta}\circ\pi_\beta(x) \text{ if } \alpha \leqslant \beta \in I \right\}.$$

Proof. Exercise. □

Observe that we topologise $\varprojlim X_\alpha$ with the subspace topology of $\prod X_\alpha$ which itself has the product topology. Hence the topology on $\varprojlim X_\alpha$ is the initial topology with respect to the canonical projections.

Proposition 1.4. *Let I be a directed set and let $(X_\alpha, f_{\alpha\beta})$ be an inverse system of nonempty compact Hausdorff spaces with index set I and let $X = \varprojlim X_\alpha$. Then X is compact Hausdorff and nonempty.*

Proof. Clearly X is Hausdorff since the T_2-axiom is closed under product and subspaces. By Tychonoff's Theorem we get that $\prod X_\alpha$ is compact. To show X is compact, it suffices to show that X is a closed subspace.

For each $\alpha \leqslant \beta \in I$, let $\Gamma_{\alpha\beta} = \{x \in \prod X_\alpha \mid \pi_\alpha(x) = f_{\alpha\beta}\circ\pi_\beta(x)\}$. Since X_α is Hausdorff, the diagonal set $D = \{(s,s) \in X_\alpha \times X_\alpha \mid s \in X_\alpha\}$ is closed. Observe that $(f_{\alpha\beta}\times \mathrm{id}_{X_\alpha})^{-1}(D)$ is closed, hence $\Gamma_{\alpha\beta}$ is closed. Furthermore,

$$X = \bigcap_{\alpha\leqslant\beta\in I} \Gamma_{\alpha\beta} \tag{1.5}$$

which proves our first assertion.

We now prove that X is nonempty. Let $n \in \mathbb{N}$ and let $\alpha_i \leqslant \beta_i \in I$ for $i = 1, \ldots, n$. We show that

$$\Gamma_{\alpha_1\beta_1} \cap \cdots \cap \Gamma_{\alpha_n\beta_n} \neq \emptyset.$$

We construct an element $(x_\alpha)_{\alpha\in I}$ that is in the aforegiven set. Since I is directed, there is $\gamma \in I$ such that $\beta_i \leqslant \gamma$ for $i = 1, \ldots, n$. Pick any $x_0 \in X_\gamma$ and put $x_\nu = f_{\nu\gamma}(x_0)$ for $\nu = \alpha_1, \beta_1, \ldots, \alpha_n, \beta_n$. Pick the rest of the $x_\alpha \in X_\alpha$ arbitrarily. Our assertion follows from Equation 1.5 and the finite intersection property. □

1.2. **Topological Groups.** A topological group G is a topological space with a compatible group structure. That is, the group operation $G\times G \to G$ sending $x\times y$ to $x\cdot y$ and inversion $G \to G$ sending x to x^{-1} are continuous maps. Observe that every group can be made into a topological group by giving it the discrete topology. For the general theory, we refer the reader text such as [Bou98] and [Mun99].

Let G be a topological group throughout. We state a few important results.

Exercise 1.3. Let C_0 be the component of the identity $e \in G$.

(1) Prove that C_0 is a closed normal subgroup of G.
(2) Prove that the component of any $x \in G$ is $xC_0 = C_0x$.
(3) Prove that G/C_0 is totally disconnected.

Exercise 1.4. Let G be compact and let $H \leqslant G$. Prove that H is open if and only if H has finite index and is closed.

Exercise 1.5. Prove that G is Hausdorff if and only if $\{e\}$ is closed in G.

1.3. **Totally Separated Spaces.** Recall that quasicomponents are defined by the equivalence relation "$x \sim y$ if there is no separation (U, V) of X with $x \in U$ and $y \in V$". It can be shown that QC_x is the intersection of all clopen neighbourhoods of x. Thus QC_x is closed. It can also be shown that $C_x \subset QC_x$ for all $x \in X$.

Recall that totally disconnected spaces are those spaces where all connected components are singletons and totally separated spaces are where all quasicomponents are singletons.

Theorem 1.6. *If X is compact Hausdorff then quasicomponents agree with the connected components. In particular, if X is compact Hausdorff then totally disconnected $\iff$ totally separated.*

Proof. It is clear that $C_x \subset QC_x$, it is enough to show that QC_x is connected. Suppose not, then there is a separation (X_1, X_2) of QC_x with $x \in X_1$. Observe that X_1 is closed in X since $X_1 = G \cap QC_x$ where G is closed in X, similarly for X_2. By the normality of X, there are open $U, V \subset X$ such that $X_1 \subset U$ and $X_2 \subset V$.

Let $\{S_\alpha\}_{\alpha \in I}$ denote the set of all clopen neighbourhoods of x. Recall that $QC_x = \cap_\alpha S_\alpha$, we claim that some $S_\alpha \subset U \cup V$. First observe that $S_\alpha \cap (X \setminus (U \cup V))$ is compact. Now,

$$\bigcap_{\alpha \in I} (S_\alpha \cap (X \setminus (U \cup V))) = QC_x \cap (X \setminus (U \cup V)) = \emptyset.$$

By the finite intersection property of compact sets, there are $\{\alpha_i\}_{i=1}^n$ such that $\cap_{i=1}^n (S_{\alpha_i} \cap (X \setminus (U \cup V))) = \emptyset$. Define $S = \cap_{i=1}^n S_{\alpha_i}$, then $S \subset U \cup V$ and S is clopen. We now show that $S \cap U$ is clopen, indeed

$$\overline{S \cap U} \subset S \cap \overline{U} = S \cap (U \cup V) \cap \overline{U} = S \cap U$$

since $V \cap \overline{U} = \emptyset$. Since $S \cap U$ is a clopen neighbourhood of x, we have $QC_x \subset S \cap U$ which forces $X_2 = \emptyset$, a contradiction. □

Henceforth, we shall use the words totally disconnected and totally separated interchangeably since we shall mainly be working in compact Hausdorff spaces.

2. Profinite Groups

2.1. **Definition.** We need a lemma before we get to the definition of profinite groups.

Lemma 2.1. *For a Hausdorff space T, the following are equivalent:*

(i) T is homeomorphic to the topological inverse limit of finite discrete spaces,
(ii) T is compact and has a clopen neighbourhood basis at each point,
(iii) T is compact and totally separated.

Proof. We first show $(i) \Rightarrow (ii)$. It follows from Proposition 1.4 that T is compact. It is given that $T = \varprojlim V_\alpha$ where V_α are finite discrete spaces. Then $\pi_\alpha^{-1}(x)$ for all $x \in V_\alpha$ form a basis for T. These are clopen since V_α have the discrete topology.

Now we show $(ii) \Rightarrow (iii)$. Let $x, y \in T$ be distinct. Since T is Hausdorff, there are open $U \ni x$ and $V \ni y$. By (ii) there is a clopen neighbourhood $x \in S \subset U$. Then $(S, X \setminus S)$ is a separation of X such that $x \in S$ and $y \in X \setminus S$, so the quasicomponents of X must be singletons.

Finally we show $(iii) \Rightarrow (i)$. Let I be the set of all equivalence relations $R \subset T \times T$ on T such that T/R is finite and discrete. This is nonempty since $T \times T \in I$. Order

I by reverse inclusion. To show that I is directed, we show that $R_1 \cap R_2 \in I$. Indeed, $R_1 \cap R_2$ is an equivalence relation and the map $\phi : T/R_1 \cap R_2 \to T/R_1 \times T/R_2$ given by $\phi = \pi_1 \times \pi_2$ is injective which gives us that $T/R_1 \cap R_2$ is finite and discrete since $T/R_1 \times T/R_2$ is finite and discrete. We claim that the canonical map $\phi : T \to \varprojlim_{R\in I} T/R$ is a homeomorphism.

First we prove that ϕ is surjective. Let $\{t_R\}_{R\in I} \in \varprojlim_{R\in I} T/R$. Then $\pi_R^{-1}(t_R)$ is nonempty and closed, hence compact. Since I is directed, finite intersections of these sets are nonempty. Our claim follows from the finite intersection property and the identity

$$\phi^{-1}\left(\{t_R\}_{R\in I}\right) = \bigcap_{R\in I} \pi_R^{-1}(t_R).$$

Now we show that ϕ is injective. Let $x, y \in T$ be distinct. Since T is totally separated, there is a clopen $U \ni x$ with $y \notin U$. Let R be the equivalence relation related to the partition U and $X \setminus U$. Then T/R is finite and discrete. Moreover, $\phi(x)$ and $\phi(y)$ will differ in the R-th slot.

Since a continuous bijection between compact Hausdorff spaces is a homeomorphism, our claim follows. □

These equivalences define a **Stone space** or **Profinite space**.

Proposition 2.2. *For a Hausdorff topological group G the following are equivalent:*

(1) G is a topological inverse limit of finite discrete groups,
(2) G is compact and e has a neighbourhood basis of clopen normal subgroups,
(3) G is compact and totally separated.

Proof. We first show $(i) \Rightarrow (iii)$. It follows from Proposition 1.4 that G is compact. Moreover, the product and subspace of totally disconnected spaces is totally disconnected, so G is totally disconnected.

We now show $(ii) \Rightarrow (i)$. Let $\mathbb{U}$ denote the set of open normal subgroups. This is nonempty since e has a basis of clopen normal subgroups. For each $U \in \mathbb{U}$, the set of cosets G/U is finite due to the compactness of G and G/U is discrete since U is open. We show that the canonical map $\phi : G \to \varprojlim_{U\in\mathbb{U}} G/U$ is an isomorphism.

To show ϕ is injective, we show that $\ker\phi$ is trivial. Indeed, for any $x \in G$ that is not the identity, there are open disjoint $V \ni e$ and $W \ni x$ since G is Hausdorff. We can find a clopen normal $U \subset V$. Moreover,

$$\ker\phi = \bigcap_{U\in\mathbb{U}} U$$

hence the kernel is trivial. We use the exact same argument as in Lemma 2.1 to show that ϕ is surjective. Since a continuous bijection between compact Hausdorff spaces is a homeomorphism, our claim follows.

Finally, we show $(iii) \Rightarrow (ii)$. By Lemma 2.1, every point has a basis of clopen sets. It suffices to show that every clopen neighbourhood of e has a clopen normal subgroup contained in it. Let U be one such neighbourhood. Set

$$V = \{v \in U \mid vU \subset U\} \qquad \text{and} \qquad H = V \cap V^{-1}.$$

We first show that V is open. Fix $v \in V$, then $uv \in U$ for each $u \in U$ and using the continuity of multiplication, we get open neighbourhoods U_u of u and V_u of v such

that $U_u V_u \subset U$. Observe that $\{U_u\}_{u \in U}$ form an open cover of U which is compact. So there are $u_1, \ldots, u_n$ such that $U_{u_1}, \ldots, U_{u_n}$ cover U. Set

$$V_v = V_{u_1} \cap \cdots \cap V_{u_n}.$$

This is an open neighbourhood of v contained in V, so V is open. Since inversion is continuous, $H = V \cap V^{-1}$ is also open. It is easy to check that H is a subgroup. By Exercise 1.4, H has finite index and is closed. The intersection of these conjugates is a clopen normal subgroup of G contained in U. □

The above equivalences define a **Profinite Group**. The intuition behind profinite groups is the same as that of a limit.

Exercise 2.1. Let $G = \varprojlim G_\alpha$ be a profinite group where G_α are finite discrete groups. Let $U_\alpha = \ker(G \to G_\alpha)$. Prove that U_α form a neighbourhood basis of the identity.

2.2. **Basic Properties.** We state a few basic properties of profinite groups like how they behave under subspaces and quotients.

Proposition 2.3. *Let G be a profinite group and $H \leqslant G$ be a closed subgroup. Then H is a profinite group.*

Proof. By Proposition 2.2, it is enough to show that H is totally disconnected and compact Hausdorff. It is clear that H has these properties since it is a subspace of a totally disconnected compact Hausdorff space G. □

Proposition 2.4. *Let G be a profinite group and $H \trianglelefteq G$ be a closed normal subgroup. Then G/H is a profinite group.*

Proof. By Proposition 2.2, it is enough to show that G/H is totally disconnected and compact Hausdorff. Let $p : G \to G/H$ denote the canonical projection map, recall that this map is open. Since H is closed, G/H is Hausdorff and G/H is compact since it is the continuous image of a compact set.

It remains to show that G/H is totally disconnected. Let $C \subset G/H$ be a connected component. Suppose $xH, yH \in C$ are distinct elements. Then $p^{-1}(xH)$ and $p^{-1}(yH)$ are disjoint and closed. By the normality of G, there are disjoint open $U \supset p^{-1}(xH)$ and $V \supset p^{-1}(yH)$, Since G is profinite, there is a clopen normal neighbourhood N of e such that $xN \subset U$ and $yN \subset V$. Observe that $xH \in p(xN)$ and since p is an open map and by Exercise 1.4, $p(xN)$ is clopen. So $C \subset p(xN)$ but $yH \notin p(xN)$, a contradiction. □

2.3. **Examples.** Profinite Groups arise naturally in mathematics. The main examples are the p-adic integers and infinite Galois groups.

Let p be a prime number. For $1 \leqslant m < n$, there is a natural map $\varphi : \mathbb{Z}/p^n\mathbb{Z} \to \mathbb{Z}/p^m\mathbb{Z}$ since $p^n\mathbb{Z} \subset p^m\mathbb{Z}$.

$$\begin{array}{ccc} \mathbb{Z}/p^n\mathbb{Z} & \xrightarrow{\varphi} & \mathbb{Z}/p^m\mathbb{Z} \\ & {\scriptstyle \pi_n} \nwarrow \quad \nearrow {\scriptstyle \pi_m} & \\ & \mathbb{Z} & \end{array}$$

This gives rise to the system

$$\cdots \longrightarrow \mathbb{Z}/p^{n+1}\mathbb{Z} \longrightarrow \mathbb{Z}/p^n\mathbb{Z} \longrightarrow \cdots \longrightarrow \mathbb{Z}/p^2\mathbb{Z} \longrightarrow \mathbb{Z}/p\mathbb{Z}.$$

This is indeed an inverse system with index set $\mathbb{N}$. We denote the inverse limit by $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$.

It turns out that $\mathbb{Z}_p$ is metrizable. We define the p-adic metric d as follows. Let $\alpha = (\alpha_n), \beta = (\beta_n) \in \mathbb{Z}_p$. If $\alpha = \beta$, set $d(\alpha, \beta) = 0$. Otherwise, there is a smallest integer n for which $\alpha_n \neq \beta_n$, then set $d(\alpha, \beta) = p^{-n}$.

3. G-modules

We shall now define a small generalisation of modules, where a group acts on an abelian group instead of a ring.

Definition 3.1. Let G be a profinite group. An **abstract G-module** is an abelian group M along with an action of G. That is a group homomorphism

$$G \xrightarrow{\sigma} \mathrm{End}_{\mathbf{Ab}}(M).$$

We shall denote $\sigma(g)(m)$ by $g \cdot m$ or simply by $g(m)$. Recall that $\mathrm{End}_{\mathbf{Ab}}(M)$ is an abelian group with the addition being pointwise.

We make two more definitions.

Definition 3.2. A **topological G-module** is an abelian topological group M with the structure of an abstract G-module where the action $G \times M \to M$ is continuous.

Definition 3.3. A **discrete G-module** is a topological G-module where M has the discrete topology.

For a subgroup $H \leqslant G$, we define the **H-invariant subgroup** as

$$M^H = \{m \in M \mid h(m) = m \text{ for all } h \in H\}.$$

Proposition 3.4. *Let G be a profinite group and let M be an topological G-module. Then the following are equivalent:*

(i) M is a discrete G-module,
(ii) For each $m \in M$, the stabilizer subgroup $G_m = \{g \in G \mid g(m) = m\}$ is open,
(iii) $M = \cup M^U$ where U runs through the open subgroups of G.

Proof. Fix an $m \in M$. Since the action $G \times M \to M$ is continuous, we can curry m to give the continuous map $\sigma : G \to M$ where $g \mapsto g(m)$. Further, M has the discrete topology, hence $G_m = \sigma^{-1}(m)$ is open. This shows $(i) \Rightarrow (ii)$. Observe that $m \in M^{G_m}$, this shows $(ii) \Rightarrow (iii)$. Let $(g, m) \in G \times M$, then there is an open subgroup U of G such that $m \in M^U$. Hence $gU \times \{m\}$ is an open neighbourhood of (g, m) which maps to $g(m)$. This shows (i). □

For discrete G-modules A and B, we can make $\mathrm{Hom}_{\mathbf{Ab}}(A, B)$ into an abstract G-module by endowing it with the action

$$g(\phi)(a) = g(\phi(g^{-1}(a))). \tag{3.5}$$

This may seem a bit peculiar, but if $\phi \in \mathrm{Hom}_{\mathbf{Ab}}(A, B)$ is a map that is fixed by G, then $g(\phi)(a) = \phi(g(a))$ i.e. ϕ is a G-module homomorphism. We can put this concretely by saying

$$\mathrm{Hom}_G(A, B) = \mathrm{Hom}_{\mathbf{Ab}}(A, B)^G.$$

This shall be of particular use to us in the next section.

4. Cohomology of Profinite Groups

Let G be a profinite group. For $n \in \mathbb{N}$ and $i = 0, \dots, n$; denote by $d_i : G^{n+1} \to G^n$ the projection

$$(\sigma_0, \dots, \sigma_n) \mapsto (\sigma_0, \dots, \widehat{\sigma_i}, \dots, \sigma_n)$$

where the hat denotes that the element has been omitted.

Let A be a discrete G-module, we form the abelian group

$$X^n(G, A) = \mathrm{Hom}_{cts}(G^{n+1}, A)$$

where the addition is pointwise. By Equation 3.5, we can make this into an abelian group. We let G act on G^{n+1} by left multiplication. Explicitly, the action is given by

$$(\sigma x)(\sigma_0, \dots, \sigma_n) = \sigma x(\sigma^{-1}\sigma_0, \dots, \sigma^{-1}\sigma_n).$$

The maps $d_i : G^{n+1} \to G^n$ induce maps, under the contravariant functor $\mathrm{Hom}(-, A)$. That is, the maps $d_i^* : X^{n-1} \to X^n$. We can now define the boundary operator $\partial^n : X^{n-1} \to X^n$ given by

$$\partial^n = \sum_{i=0}^{n} (-1)^i d_i^*.$$

For $x \in X^{n-1}$, we get the function

$$(\partial^n x)(\sigma_0, \dots, \sigma_n) = \sum_{i=0}^{n} (-1)^i x(\sigma_0, \dots, \widehat{\sigma_i}, \dots, \sigma_n)$$

For $n = 0$, $\partial^0 : A \to X^0$ associates a to the function $(\sigma \mapsto a)$.

Proposition 4.1. *The sequence*

$$0 \longrightarrow A \xrightarrow{\partial^0} X^0 \xrightarrow{\partial^1} X^1 \xrightarrow{\partial^2} X^2 \xrightarrow{\partial^3} \cdots \tag{4.2}$$

is exact.

Proof. We first show that this is a complex, i.e. $\partial^2 = 0$. Firstly,

$$\partial^1 \circ \partial^0(a)(\sigma_0, \sigma_1) = (\partial^0 a)(\sigma_1) - (\partial^0 a)(\sigma_0) = a - a = 0.$$

Now for $n \geqslant 2$ and $x \in X^{n-2}$,

$$\partial^n \circ \partial^{n-1}(x)(\sigma_0, \dots, \sigma_n) = \sum_{i=0}^{n} (-1)^i (\partial^{n-1} x)(\sigma_0, \dots, \widehat{\sigma_i}, \dots, \sigma_n)$$

$$= \sum_{i=0}^{n} (-1)^i \sum_{j=0}^{n-1} (-1)^j \begin{cases} x(\sigma_0, \dots, \widehat{\sigma_i}, \dots, \widehat{\sigma_{j-1}}, \dots, \sigma_n) & j > i \\ x(\sigma_0, \dots, \widehat{\sigma_j}, \dots, \widehat{\sigma_i}, \dots, \sigma_n) & \text{otherwise} \end{cases}.$$

Since each term appears twice, it will have the coefficients $(-1)^{i+j}$ and $(-1)^{i+j-1}$, so they cancel and we get 0.

Now for the exactness, we define $D^n : X^{n+1} \to X^n$. Firstly, $D^{-1} : X^0 \to A$ sends x to $x(1)$. For $x \in X^{n+1}$,

$$(D^n x)(\sigma_0, \dots, \sigma_n) = x(1, \sigma_0, \dots, \sigma_n).$$

It should be noted that D^n are not G-module homomorphisms and are only homomorphisms of abelian groups. Some calculation shows that, for $n \geqslant 0$,

$$D^n \circ \partial^{n+1} + \partial^n \circ D^{n-1} = id.$$

Since $\partial^2 = 0$, we have $\operatorname{Im}\partial^n \subset \ker\partial^{n+1}$. Evaluating the above identity at $x \in \ker\partial^{n+1}$, we get $\partial^n(D^{n-1}x) = x$, so $x \in \operatorname{Im}\partial^n$. Hence $\ker\partial^{n+1} \subset \operatorname{Im}\partial^{n-1}$, and the sequence is exact. □

Exercise 4.1. For $n \geqslant 0$, show that $D^n \circ \partial^{n+1} + \partial^n \circ D^{n-1} = id$. (The definition of D^n can be found in Proposition 4.1)

We now define the endofunctor "fixed module" on G-**Mod**. It maps A to A^G and maps morphisms $\varphi : A \to B$ to the restriction maps $\varphi : A^G \to B^G$ (abusing notation). We now define the (homogeneous) n-**cochains** of G with coefficients in A as

$$C^n(G, A) = X^n(G, A)^G.$$

Applying the fixed module functor to Equation 4.2, we get

$$C^0(G, A) \xrightarrow{\partial^1} C^1(G, A) \xrightarrow{\partial^2} C^2(G, A) \xrightarrow{\partial^3} \cdots,$$

observe that the left exactness is lost but is still a complex, i.e. $\partial^2 = 0$. We now set

$$Z^n(G, A) = \ker(C^n(G, A) \xrightarrow{\partial^{n+1}} C^{n+1}(G, A))$$

$$B^n(G, A) = \operatorname{Im}(C^{n-1}(G, A) \xrightarrow{\partial^n} C^n(G, A))$$

with $B^0(G, A) = 0$. These are called the (homogeneous) n-**cocycles** and (homogeneous) n-**coboundaries** respectively. And finally define the n-**dimensional cohomology group** as

$$H^n(G, A) = Z^n(G, A)/B^n(G, A).$$

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